# Numerical methods

## Lecture 3: Application of GE, LU decomposition

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# Applications of Gaussian elimination

- Solving SLE (we had an example)
- Calculating the determinant: since the steps of GE don't change the determinant

$$\det(A^{(0)}) = \det(A^{(n-1)}) = \prod_{k=1}^{n} a_{kk}^{(k-1)}$$

  Careful upon permuting rows or columns (det changes)!
- Solving SLE for severel right-hand-side vectors ($b$): all at once, so the matrix is handled once.

$$[A|b_1|b_2|b_3] \to \text{GE \& back subs.} \to [I|x_1|x_2|x_3]$$

# Applications of Gaussian elimination

- Matrix inverse: solve $A \cdot X = I$.

$$A \cdot [x_1 | \dots | x_n] = [e_1 | \dots | e_n] \quad \Longleftrightarrow \quad \begin{aligned} Ax_1 &= e_1 \\ &\vdots \\ Ax_n &= e_n \end{aligned}$$

Reduced to the previous item.

$$[A \,|\, I] \to \text{GE \& back subs.} \to [I \,|\, A^{-1}],$$

(Remark. Switching rows does not change the inverse, switching columns does.)

# Inverse example

**Example:** determinant and inverse with GE

What is the determinant and the inverse of the below matrix?
(c.f. Lecture 2)

$$\begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix}$$

$$\det(A) = \det(A^{(2)}) = \begin{vmatrix} 2 & 0 & 3 \\ 0 & 5 & 4 \\ 0 & 0 & -1 \end{vmatrix} = 2 \cdot 5 \cdot (-1) = -10$$

**Elimination (forward)**:
**Step 1:**

$$\text{Row 2} := \text{Row 2} \underbrace{-\left(\frac{-4}{2}\right)}_{+2} \cdot \text{Row 1}$$

$$\text{Row 3} := \text{Row 3} \underbrace{-\left(\frac{6}{2}\right)}_{-3} \cdot \text{Row 1}$$

$$\left[\begin{array}{rrr|rrr} 2 & 0 & 3 & 1 & 0 & 0 \\ -4 & 5 & -2 & 0 & 1 & 0 \\ 6 & -5 & 4 & 0 & 0 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{rrr|rrr} 2 & 0 & 3 & 1 & 0 & 0 \\ 0 & 5 & 4 & 2 & 1 & 0 \\ 0 & -5 & -5 & -3 & 0 & 1 \end{array}\right] \rightsquigarrow$$

**Step 2:**

$$\text{Row 3} := \text{Row 3} \underbrace{- \left(\frac{-5}{5}\right)}_{+1} \cdot \text{Row 2}$$

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 3 & 1 & 0 & 0 \\ 0 & 5 & 4 & 2 & 1 & 0 \\ 0 & -5 & -5 & -3 & 0 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|ccc} 2 & 0 & 3 & 1 & 0 & 0 \\ 0 & 5 & 4 & 2 & 1 & 0 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{array}\right] \rightsquigarrow$$

**Back substitution (backward)**:

Row 3 $:=$ Row 3 $/(-1)$
Row 2 $:=$ Row 2 $-$ 4 $\cdot$ new Row 3
Row 1 $:=$ Row 1 $-$ 3 $\cdot$ new Row 3

$$\left[\begin{array}{rrr|rrr} 2 & 0 & 3 & 1 & 0 & 0 \\ 0 & 5 & 4 & 2 & 1 & 0 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{rrr|rrr} 2 & 0 & 0 & -2 & 3 & 3 \\ 0 & 5 & 0 & -2 & 5 & 4 \\ 0 & 0 & 1 & 1 & -1 & -1 \end{array}\right] \rightsquigarrow$$

Row 2 := Row 2 /5
Row 1 := Row 1 − 0 · Row 2

Row 1 := Row 1 /2.

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & -2 & 3 & 3 \\ 0 & 5 & 0 & -2 & 5 & 4 \\ 0 & 0 & 1 & 1 & -1 & -1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & \frac{3}{2} & \frac{3}{2} \\ 0 & 1 & 0 & -\frac{2}{5} & 1 & \frac{4}{5} \\ 0 & 0 & 1 & 1 & -1 & -1 \end{array}\right] = [I|A^{-1}]$$

The inverse is on the right-hand-side.

# Remarks on Gaussian elimination

Is the SLE solvable at all? Should we check?
*We'll find out during GE.*

It is solvable, but we might not be able to finish with GE?
*May happen...* $\rightsquigarrow$ *pivoting*

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot x = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Pivoting: switching rows and/or columns.

**Definition:** partial pivoting

In step $k$ choose an index $m$, such that $\left|a_{mk}^{(k-1)}\right|$ is maximal ($m \in \{ k, k+1, \ldots, n \}$), and then switch rows $k$ and $m$.

**Definition:** full pivoting

In step $k$ choose a pair of indices $(m_1, m_2)$, such that $\left|a_{m_1 m_2}^{(k-1)}\right|$ is maximal ($m_1, m_2 \in \{ k, k+1, \ldots, n \}$), and then switch rows $k$ and $m_1$, and columns $k$ and $m_2$.

Switching rows does not alter the solution.
Switching columns results in the change of the coordinates of the solution accordingly.

**Theorem**

The GE can be carried out without switching rows or columns
$\iff \quad a_{kk}^{(k-1)} \neq 0 \ (k = 1, 2, \ldots, n-1)$.

**Proof.** trivially by the steps of GE. $\square$

**Definition:** principal minors

The principal minors of $A$ are the determinants

$$D_k = \det\left(\begin{bmatrix} a_{11} & \ldots & a_{1k} \\ \vdots & & \vdots \\ a_{k1} & \ldots & a_{kk} \end{bmatrix}\right), \quad (k = 1, 2, \ldots, n)$$

i.e. the determinants of the upper-left $k \times k$ submatrices of $A$.

**Theorem**

$D_k \neq 0 \ (k = 1, 2, \ldots, n-1) \quad \Leftrightarrow \quad a_{kk}^{k-1} \neq 0 \ (k = 1, 2, \ldots, n-1).$

**Proof.** The steps of the GE preserve the determinant, so

$$D_k = a_{11} \cdot a_{22}^{(1)} \cdot \ldots \cdot a_{kk}^{(k-1)} = D_{k-1} \cdot a_{kk}^{(k-1)}.$$

Remark. $D_n \neq 0$ and $a_{nn}^{(n-1)} \neq 0$ is not required for the GE, just for the unique solvability of the SLE. $\square$

**Remarks:**

- If the SLE is solvable, then partial pivoting is sufficient.
- Numerically it is better to apply pivoting. (It makes the divisions cause less errors.)
- Switching rows and/or columns alter the determinant!

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# Operation count of GE

**Theorem:** Operation count of GE

$$\frac{2}{3}n^3 + \mathcal{O}(n^2)$$

**Proof.** For a fixed $k$, from the formula of the step $k$

$$a_{ij}^{(k)} = a_{ij}^{(k-1)} - \frac{a_{ik}^{(k-1)}}{a_{kk}^{(k-1)}} \cdot a_{kj}^{(k-1)} \qquad \begin{aligned} &k = 1, \ldots, n-1; \\ &i = k+1, \ldots, n; \\ &j = k+1, \ldots, n, n+1. \end{aligned}$$

$(n-k)$ divisions, $(n-k)(n-k+1)$ multiplications and $(n-k)(n-k+1)$ additions are needed.
Altogether $(n-k)(2(n-k)+3)$ operations.

$s := n - k$

$$\sum_{k=1}^{n-1}(n-k)(2(n-k)+3) = \sum_{s=1}^{n-1} s(2s+3) = 2\sum_{s=1}^{n-1} s^2 + 3\sum_{s=1}^{n-1} s =$$
$$= 2\frac{(n-1)n(2n-1)}{6} + 3\frac{(n-1)n}{2} = \frac{2}{3}n^3 + \mathcal{O}(n^2). \quad \square$$

**Remark:** $f$ is an $\mathcal{O}(n^2)$ function, if $\frac{f(n)}{n^2}$ is bounded for all $n \in \mathbb{N}$-re. (In our case $f$ is of the form $a \cdot x^2 + b \cdot x + c$.)

# Operation count of back substitution

Operation count of solving a triangular SLE

**Theorem:** Operation count of back substitution

$$n^2 + \mathcal{O}(n)$$

**Proof.**

$$x_n = \frac{a_{nn+1}^{(n-1)}}{a_{nn}^{(n-1)}}, \quad x_i = \frac{1}{a_{ii}^{(i-1)}} \left( a_{in+1}^{(i-1)} - \sum_{j=i+1}^{n} a_{ij}^{(i-1)} \cdot x_j \right) \quad (i = n-1, \dots, 1).$$

For the $i$th row 1 division, $(n-i)$ multiplications and $(n-i)$ addition. Alltogether $2(n-i)+1$ operations. $(:= n-i)$.

$$1 + \sum_{s=1}^{n-1} (2s+1) = 1 + 2 \cdot \frac{n(n-1)}{2} + (n-1) = n^2 + \mathcal{O}(n). \quad \square$$

# Left multiplication with lower triangular matrices

What happens, if we multiple a matrix $A \in \mathbb{R}^{3\times 3}$ with the below matrix $L \in \mathbb{R}^{3\times 3}$ from the left?

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}$$

Row 2 $:=$ Row 2 $+ 2 \cdot$ Row 1.

# Left multiplication with lower triangular matrices

What happens if we multiply a matrix $A \in \mathbb{R}^{3\times 3}$ with the below matrix $L \in \mathbb{R}^{3\times 3}$ from the left?

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

Row 2 := Row 2 $+2\cdot$ Row 1
Row 3 := Row 3 $-3\cdot$ Row 1
($\sim$ Step 1 of GE)

# Steps of the GE with matrix multiplication

Let us write step $k$ of GE the same way! ($A \in \mathbb{R}^{n\times n}$)

$$L_k = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & -l_{k+1k} & 1 & & \\ & & \vdots & & \ddots & \\ & & -l_{nk} & & & 1 \end{pmatrix} = I - \ell_k e_k^\top, \quad \ell_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ l_{k+1k} \\ \vdots \\ l_{nk} \end{pmatrix}.$$

(The zero elements are not shown in $L_k$.)

So if $l_{ik} = \dfrac{a_{ik}^{(k-1)}}{a_{kk}^{(k-1)}} \quad (k = 1, \dots, n-1;\;\; i = k+1, \dots, n)$,

then $L_k \cdot A^{(k-1)} = A^{(k)}$, so we arrived at step $k$ of GE.

**Example:** GE with the $L_k$ matrices

Let us write the steps of GE using matrix multiplications for the below matrix (the same as before)!

$$A = \begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix}$$

**Solution.** **Step 1**

$$A^{(1)} = L_1 \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 5 & 4 \\ 0 & -5 & -5 \end{bmatrix}$$

**Step 2**

$$A^{(2)} = L_2 \cdot A^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 3 \\ 0 & 5 & 4 \\ 0 & -5 & -5 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 5 & 4 \\ 0 & 0 & -1 \end{bmatrix} =: U$$

So $A^{(2)} = L_2 \cdot L_1 \cdot A =: U$.
$U$ denotes the upper triangular form.

Express $A$:

$$A = \underbrace{L_1^{-1} \cdot L_2^{-1}}_{=:L} \cdot U = L \cdot U.$$

Thus we arrived at the $LU$ decomposition of $A$, discussed soon...

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**Definition:** lower triangular matrix

The matrix $L \in \mathbb{R}^{n \times n}$ is called a *lower triangular matrix*, if $l_{ij} = 0$ for $i < j$. (All zeros above the diagonal.)

$$\mathcal{L} := \{\, L \in \mathbb{R}^{n \times n} \,:\, l_{ij} = 0 \; (i < j) \,\},$$
$$\mathcal{L}_1 := \{\, L \in \mathbb{R}^{n \times n} \,:\, l_{ij} = 0 \; (i < j), \;\; l_{ii} = 1 \,\}.$$

**Definition:** upper triangular matrix

The matrix $U \in \mathbb{R}^{n \times n}$ is called an *upper triangular matrix*, if $u_{ij} = 0$ for $i > j$. (All zeros below the diagonal.)

$$\mathcal{U} := \{\, U \in \mathbb{R}^{n \times n} \,:\, u_{ij} = 0 \; (i > j) \,\},$$
$$\mathcal{U}_1 := \{\, U \in \mathbb{R}^{n \times n} \,:\, u_{ij} = 0 \; (i > j), \;\; u_{ii} = 1 \,\}.$$

**Proposition:** about triangular matrices

1. If $L', L'' \in \mathcal{L}$, then $L' \cdot L'' \in \mathcal{L}$.
2. If $U', U'' \in \mathcal{U}$, then $U' \cdot U'' \in \mathcal{U}$.
3. If $L', L'' \in \mathcal{L}_1$, then $L' \cdot L'' \in \mathcal{L}_1$.
4. If $U', U'' \in \mathcal{U}_1$, then $U' \cdot U'' \in \mathcal{U}_1$.
5. If $L \in \mathcal{L}$ and $\exists L^{-1}$, then $L^{-1} \in \mathcal{L}$.
6. If $U \in \mathcal{U}$ and $\exists U^{-1}$, then $U^{-1} \in \mathcal{U}$.
7. If $L \in \mathcal{L}_1$, then $\exists L^{-1}$ and $L^{-1} \in \mathcal{L}_1$.
8. If $U \in \mathcal{U}_1$, then $\exists U^{-1}$ and $U^{-1} \in \mathcal{U}_1$.

**Proof.** Homework.
(Not difficult. Optional. May be submitted on practice.) $\square$

**Definition:** $L_k$

$L_k := I - \ell_k e_k^\top \in \mathbb{R}^{n\times n}$, with $\ell_k \in \mathbb{R}^n$, $(\ell_k)_i = 0$ $(i \leq k)$ and $e_k \in \mathbb{R}^n$ the $k$th canonical unit vector.

**Proposition:** The inverse of $L_k$

$$L_k^{-1} = I + \ell_k e_k^\top .$$

**Proof:**

$$L_k \cdot L_k^{-1} = (I - \ell_k e_k^\top)(I + \ell_k e_k^\top) = I \underbrace{- \ell_k e_k^\top + \ell_k e_k^\top}_{0} - \ell_k \underbrace{e_k^\top \ell_k}_{0} e_k^\top = I. \quad \square$$

Visually?

# About the $L_k$ matrices

How are two such matrices multiplied?

$$\begin{array}{cc} & \begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 3 & 1\end{pmatrix} \\ \begin{pmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 1 & 0 & 1\end{pmatrix} & \begin{pmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 1 & 3 & 1\end{pmatrix}\end{array}$$

$$\begin{array}{cc} & \begin{pmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 1 & 0 & 1\end{pmatrix} \\ \begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 3 & 1\end{pmatrix} & \begin{pmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 7 & 3 & 1\end{pmatrix}\end{array}$$

Easy on the left-hand-side. In general too.

**Proposition:** product of $L_k$ matrices

$$L_1^{-1} \cdot L_2^{-1} \cdots L_{n-1}^{-1} = I + \ell_1 e_1^\top + \ell_2 e_2^\top + \ldots + \ell_{n-1} e_{n-1}^\top.$$

Visually?

**Proof.** By induction.

- $$\begin{aligned} L_1^{-1} \cdot L_2^{-1} &= (I + \ell_1 e_1^\top)(I + \ell_2 e_2^\top) = \\ &= I + \ell_1 e_1^\top + \ell_2 e_2^\top + \ell_1 \underbrace{(e_1^\top \ell_2)}_{0} e_2^\top = \\ &= I + \ell_1 e_1^\top + \ell_2 e_2^\top \end{aligned}$$

- Assume that $k+1 \le n-1$ and

$$L_1^{-1} \cdot L_2^{-1} \cdots L_k^{-1} = I + \ell_1 e_1^\top + \ell_2 e_2^\top + \ldots + \ell_k e_k^\top .$$

- $L_1^{-1} \cdot L_2^{-1} \cdots L_k^{-1} \cdot L_{k+1}^{-1} =$

$$= (I + \ell_1 e_1^\top + \ell_2 e_2^\top + \ldots + \ell_k e_k^\top)(I + \ell_{k+1} e_{k+1}^\top) =$$

$$= I + \ell_1 e_1^\top + \ell_2 e_2^\top + \ldots + \ell_k e_k^\top + \ell_{k+1} e_{k+1}^\top +$$

$$+ \underbrace{\ell_1 e_1^\top \ell_{k+1} e_{k+1}^\top + \ldots + \ell_k e_k^\top \ell_{k+1} e_{k+1}^\top}_{\text{cancel}} =$$

$$= I + \ell_1 e_1^\top + \ell_2 e_2^\top + \ldots + \ell_k e_k^\top + \ell_{k+1} e_{k+1}^\top = \checkmark . \qquad \square$$

**Definition:** $LU$ decomposition

The product $L \cdot U$ is called the $LU$ decomposition of the matrix $A$, if

$$A = LU, \quad L \in \mathcal{L}_1, \;\; U \in \mathcal{U}.$$

# $LU$ decomposition with Gaussian elimination

GE can be written using lower triangular matrices:

$$L_{n-1} \cdots L_2 \cdot L_1 \cdot A = U,$$

then multiply from the inverses:

$$A = \underbrace{L_1^{-1} \cdot L_2^{-1} \cdots L_{n-1}^{-1}}_{L} \cdot U = LU.$$

The above product is also a lower triangular matrix.

The operation count is the same as for the GE. (Since we did the same thing.)

**Example:** $LU$ decomposition with GE

Find the $LU$ decomposition of the below matrix (like before)

$$A = \begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix}$$

- **a** writing the details for the $L_k$ matrices and the calculation,
- **b** and then in a 'compact' way!

**Solution: (a)** **Step 1**

$$A^{(1)} = L_1 \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 5 & 4 \\ 0 & -5 & -5 \end{bmatrix}$$

$$L_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

**Step 2**

$$A^{(2)} = L_2 \cdot A^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 3 \\ 0 & 5 & 4 \\ 0 & -5 & -5 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 5 & 4 \\ 0 & 0 & -1 \end{bmatrix} =: U$$

$$L_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & {\color{red}-1} & 1 \end{bmatrix}$$

So $A^{(2)} = L_2 \cdot L_1 \cdot A =: U$

Express $A$:

$$A = \underbrace{L_1^{-1} \cdot L_2^{-1}}_{=:L} \cdot U = L \cdot U.$$

So $L = L_1^{-1} \cdot L_2^{-1}$. Only copying, no actual multiplication needed.

$$L = L_1^{-1} \cdot L_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}$$

Verify!

**(b) Compactly:**

$$\begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix} \rightsquigarrow \left[\begin{array}{r|rr} 2 & 0 & 3 \\ \hline -2 & 5 & 4 \\ 3 & -5 & -5 \end{array}\right] \rightsquigarrow \left[\begin{array}{r|r|r} 2 & 0 & 3 \\ \hline -2 & 5 & 4 \\ \cline{2-3} 3 & -1 & -1 \end{array}\right]$$

To read the solution:

$$A = \begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 3 \\ 0 & 5 & 4 \\ 0 & 0 & -1 \end{bmatrix} = L \cdot U$$

**Theorem:** Existence of $LU$ decomposition

If the Gaussian elimination can be carried out without switching rows or columns (i.e. $a_{kk}^{(k-1)} \neq 0$ $(k = 1, \ldots, n-1)$), then the $LU$ decomposition of the matrix exists.

**Proof.** That's what we did... One can write the $L_k$ matrices, and form $L$ and $U$. □

**Remarks:**

- $u_{kk} = a_{kk}^{(k-1)}$ and $D_k = a_{11} \cdot a_{22}^{(1)} \cdots a_{kk}^{(k-1)}$
- If $A$ has an $LU$ decomposition, then the diagonal elements of $U$ are not zeros, i.e. $u_{kk} = a_{kk}^{(k-1)} \neq 0$.
- $a_{nn}^{(n-1)} \neq 0 \;\Leftrightarrow\; \det(A) = D_n \neq 0$.
- If GE can be done, but $a_{nn}^{(n-1)} = 0$, then the $LU$ decomposition exists, but $\det(A) = \det(L) \cdot \det(U) = \det(U) = 0$. The SLE does not have a unique solution (none or infinitely many).

**Theorem:** Existence and uniqueness of $LU$ decomposition

- If $D_k \neq 0 \ \ (k = 1, \ldots, n-1)$, then the $LU$ decomposition of $A$ exists and $u_{kk} \neq 0 \ \ (k = 1, \ldots, n-1)$.
- If furthermore $\det(A) \neq 0$, then the decomposition is unique.

**Proof. Existence.** Follows from the previous theorems. The GE can be carried out, the elements of $L$ and $U$ can be formed.

**Uniqueness.** Indirect. Assume two distinct $LU$ decompositions

$$A = L_1 \cdot U_1 = L_2 \cdot U_2.$$

$$A = L_1 \cdot U_1 = L_2 \cdot U_2.$$

Multiply with the inverses:

$$U_1 \cdot U_2^{-1} = L_1^{-1} \cdot L_2.$$

The inverses exist, since

$$\det(A) = \det(L_i) \cdot \det(U_i) = \det(U_i) \neq 0. \qquad (i = 1, 2)$$

left-hand-side: upper triangular matrix,
right-hand-side: lower triangular matrix with 1 diagonal.

$$\begin{aligned} U_1 \cdot U_2^{-1} = I \quad &\Longrightarrow \quad U_1 = U_2, \\ L_1^{-1} \cdot L_2 = I \quad &\Longrightarrow \quad L_1 = L_2. \end{aligned}$$

Contradiction! $\square$

**Remark:** $L$ and $U$ with GE

The above calculations for the $LU$ decomposition of $A$ can be summarized as follows:

$$L \in \mathcal{L}_1 \text{ és } l_{ij} = \frac{a_{ij}^{(j-1)}}{a_{jj}^{(j-1)}} \ (i > j), \qquad U \in \mathcal{U} \quad \text{és} \quad u_{ij} = a_{ij}^{(i-1)} \ (i \leq j).$$

# Why is the $LU$ decomposition useful?

Assume that

- the SLE $Ax = b$ LER has a solution,
- we have the $A = LU$ decomposition.

Then instead of $Ax = L \cdot \underbrace{U \cdot x}_{y} = b$ solve the systems

$(\frac{2}{3}n^3 + \mathcal{O}(n^2))$

1. $Ly = b$ (lower triangular) and $(n^2 + \mathcal{O}(n))$
2. $Ux = y$ (upper triangular). $(n^2 + \mathcal{O}(n))$

For comparison, for a matrix-vector product:
$n \cdot (2n - 1) = 2n^2 + \mathcal{O}(n)$.

Of course the $LU$ decomposition must be formed. $(\frac{2}{3}n^3 + \mathcal{O}(n^2))$
Advantageous if $A$ is the same for many SLE.

# Examples in Matlab

1. $LU$ decomposition for 'smaller' matrices ($n \approx 7$),
2. and 'bigger' matrices ($n \approx 50$) with colors.
3. Solving SLE using $LU$ decomposition.
4. Runtime comparison for solving many ($m \approx 10, 100$) big ($n \approx 50, 100, 200$) SLE: GE vs. using $LU$ decomposition.